physical system in near real-time. This captures the dynamic nature of operational environments.

Video data can be integrated into a digital twin using three main strategies:

CONTINUOUS STREAMING

The simplest, but most costly method. The entire video feed is continuously transmitted to the digital twin for real-time monitoring and storage. This demands high bandwidth and storage resources.

CHANGE DETECTION

A more efficient variation where data is only sent or stored when a relevant physical change (e.g., movement or new appearance) is detected in the video stream. This is common in security applications where only event-based fragments are transmitted.

EDGE PROCESSING WITH SELECTIVE TRANSMISSION

The most resource-efficient method. Raw video streams consume massive bandwidth when transmitted to remote servers or the cloud. Instead, video data is processed locally at the edge (close to the physical asset). Only derived, processed data (for example, 'vehicle detected', 'occupancy count is 3') is selectively transmitted, drastically reducing bandwidth and improving efficiency.

> ***NOTE***
>
> *At the edge* refers to processing data and running analytics close to where the physical asset or data is located, rather than transporting the data to a centralized data center or the cloud. It offers benefits in terms of reduced latency, bandwidth efficiency, and the ability to respond to data quickly.